EE6204 Systems Analysis

Beginner Notes: Week 1 – Linear Programming Foundations

Living course note

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Scope and sources. This note covers Week 1 only: the introduction to linear programming, problem formulation, the geometry of two-variable LPs, and conversion toward standard form. It uses `resources/lecture-01-04-linear-and-nonlinear-programming.pdf` (Lecture 1), `resources/introduction-to-linear-programming.pdf`, and `resources/week-01-transcript.txt`. The slides control formal notation; the transcript is used for teaching emphasis.

Contents

1 Optimisation: choosing the best feasible action

An optimisation problem is a structured way to make a decision. We choose values for *decision variables*, measure their quality using an *objective function*, and enforce limits using *constraints*. In a general minimisation problem,

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad x \in \mathcal{F},$$

where $\mathcal{F}$ is the *feasible set*: the choices that satisfy every constraint. A numerical answer is not a valid solution if it violates a capacity, demand, safety, or non-negativity condition.

1.1 How to formulate a real problem

Start in words and make each modelling choice explicit.

1. Define each variable, including units: for example, "x = number of suits produced per week."
2. State whether the objective is to maximise a benefit (profit, throughput) or minimise a cost, time, or waste.
3. Translate each physical condition into an equation or inequality.

4. State the domain. Production amounts usually require $x \geq 0$; whole objects may require $x \in \mathbb{Z}$.

Check signs against the wording: "at most" gives $\leq$; "at least" gives $\geq$. A cost-minimisation model must also include a reason to produce. Otherwise, producing nothing can be the correct mathematical minimum but the wrong operational model.

2 Linear programming (LP)

An LP has a linear objective and linear constraints. A standard maximisation form is

$$\max\ c^{\mathsf{T}}x \quad \text{subject to} \quad Ax \leq b, \qquad x \geq 0.$$

"Linear" means a variable appears only to the first power with a constant coefficient. Terms such as x_1x_2, x_1^2, or x_1/x_2 make a model nonlinear.

2.1 Worked formulation: product mix

Suppose x suits and y gowns earn 30 and 50 units of profit. If material stocks impose three resource limits, the model is

$$\begin{aligned} \max \quad & 30x + 50y \\ \text{s.t.} \quad & 2x + y \leq 16, \\ & x + 2y \leq 11, \\ & x + 3y \leq 15, \\ & x, y \geq 0. \end{aligned}$$

Every coefficient must have a physical interpretation. In the first constraint, 2 is the amount of the first material used per suit, 1 the amount per gown, and 16 the available amount. The last two inequalities exclude negative production; they are constraints too.

3 The graphical view of an LP

With two variables, a linear equality is a straight line and a linear inequality keeps one side of that line. The feasible region is the intersection of all allowed half-planes. To draw it, replace each inequality by equality to obtain a boundary, use a test point to choose the permitted side, and retain only points satisfying every constraint.

The objective has level lines $c^{\mathsf{T}}x = z$. Slide such a line in the improving direction until it makes its last contact with the feasible region. For a finite optimum, a corner (also called an extreme point) attains an optimum. This is why evaluating feasible corners solves a two-variable LP.

Recognise three special outcomes:

- **Infeasible:** no point satisfies all constraints simultaneously.
- **Unbounded:** one can improve the objective forever along a feasible direction.
- **Multiple optima:** the objective line is parallel to an optimal feasible edge, so every point on that edge has the same best value.

Graphing gives valuable intuition but does not scale to many variables, where the feasible set is a high-dimensional polyhedron.

4 Preparing an LP for systematic solution

Simplex methods use equality constraints and nonnegative variables. The following conversions preserve the feasible choices when the new variables are included:

$$
\begin{aligned}
a_i^\top x \le b_i &\implies a_i^\top x + s_i = b_i, \quad s_i \ge 0 && \text{(slack variable)},\\
a_i^\top x \ge b_i &\implies a_i^\top x - e_i = b_i, \quad e_i \ge 0 && \text{(surplus variable)},\\
x_k \text{ free} &\implies x_k = x_k^+ - x_k^-, \quad x_k^+, x_k^- \ge 0.
\end{aligned}
$$

A slack variable measures unused capacity: if $2x + y + s = 16$, then $s = 0$ means the resource is fully used. A surplus variable measures the amount by which a lower-bound requirement is exceeded. These are not arbitrary algebraic tricks; they give an inequality's missing amount an explicit nonnegative meaning.

5 Week 1 learning checklist

- Can I identify variables, objective, constraints, and domains in a word problem?
- Can I turn "at most" and "at least" into correctly signed inequalities?
- Can I draw a two-variable feasible region and diagnose infeasibility, unboundedness, or multiple optima?
- Can I add slack or surplus variables and explain their physical meaning?

Source: `resources/lecture-01-04-linear-and-nonlinear-programming.pdf`*, Lecture 1;* `resources/introduction-to-linear-programming.pdf`*, pp. 1–28;* `resources/week-01-transcript.txt`*..*